

Dr. Vicente Valenzuela-Villaseca

Lawrence Postdoctoral Fellow
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Summary

I am a plasma physicist with broad interests in laboratory astrophysics and astrophysical theory. My research develops and uses pulsed-power and laser-driven platforms, supported by laser, self-emission, and particle diagnostics, together with analytic theory and numerical simulations, to study fundamental processes relevant to astrophysical plasmas. My work encompasses angular momentum transport, particle acceleration and heating in collisionless shocks and magnetic reconnection, MHD and kinetic instabilities, turbulence, MHD shocks, and the structure of plasma jets.

Employment

- 2025–Present **Lawrence Postdoctoral Fellow**, *National Ignition Facility and Photon Science*, Lawrence Livermore National Laboratory, USA.
- 2022–2025 **Postdoctoral Research Associate**, *Department of Astrophysical Sciences*, Princeton University, USA.

Visiting Appointments and Fellowships

- 2025–Present **Visiting Scientist**, *Plasma Science and Fusion Center*, Massachusetts Institute of Technology, USA.
- 2025–Present **Sigma Xi**, *Full Member (by invitation)*, USA.
- 2023–Present **Academic Visitor**, *Plasma Physics Group*, Imperial College London, UK.
- 2017–Present **Fellow of the Royal Astronomical Society**, *London*, UK.

Education

- 2022 **Imperial College London**, *Ph. D. in Plasma Physics*,
Thesis: “Experimental study of magneto-inertially driven, differentially rotating plasma flows”.
Advisor: Prof. Sergey V. Lebedev.
- 2018 **Pontificia Universidad Catolica de Chile**, *M. Sc. in Experimental Physics*,
Summa Cum Laude,
Thesis: “Experiments on laser-produced annular plasmas”.
Advisor: Prof. Mario Favre.
- 2015 **Pontificia Universidad Catolica de Chile**, *B. Sc. in Physics*,
Summa Cum Laude,
Thesis: “Experiments on conical wire array Z-pinches”.
Advisor: Dr. Felipe Veloso.

Funding, Awards, Scholarships

Research grants

- 2026–2028 **Principal Investigator**, *Investigating the instability of black hole accretion discs with differentially rotating plasma experiments driven by energetic lasers*, Laboratory Directed Research and Development grant, Department of Energy (LDRD-DoE), USA, (\$415k/year, \$1.2M total).

Funded proposals

- 2026–2027 **Principal Investigator**: *Evolution of magnetized differentially rotating plasmas driven by long-pulse lasers*, 2 full days of beam time at the Omega-60 laser facility, National Laser Users’ Facility (NLUF) program, USA (Estimated equivalent facility-access value: ~\$100k per OMEGA-60 shot day, plus \$18k for target fabrication).

- 2027 **Principal Investigator** (Co-PI: Julien Fuchs, CNRS, France): *Investigation of the non-linear evolution of long-lived perpendicular magnetized shocks*, 1 full day of beam time at the Omega-EP laser facility, National Laser Users' Facility (NLUF) program, USA. (Estimated equivalent facility-access value: ~\$100k per OMEGA-EP shot day)
- 2026–2027 **Co-Principal Investigator** (PI: Derek Schaeffer, UCLA, USA): *Particle heating by strongly-driven shocks in magnetized HED plasmas*, 2 full days of beam time at the Omega-60 laser facility, National Laser Users' Facility (NLUF) program, USA (Estimated equivalent facility-access value: ~\$100k per OMEGA-60 shot day, plus \$18k for target fabrication).
- 2024–2025 **Principal Investigator**: *Magnetic Ω -Effect in Differentially Rotating Plasma Flows*, 2 full days of beam time at the Omega-60 laser facility, National Laser Users' Facility (NLUF) program, USA (Estimated equivalent facility-access value: ~\$100k per OMEGA-60 shot day, plus \$14k for target fabrication).
- 2018 **Postgraduate Research Grant**: *New Laboratory Astrophysics Experiments Looking at Magnetically Driven Rotating Plasmas*, Royal Astronomical Society, UK (\$1.5k).
- 2013 **Undergraduate Research Grant**: *Experiments on Plasma Jets with Potential Interest in Astrophysical Scaling*, Pontificia Universidad Catolica de Chile, Chile (\$200).

Awards

- 2024 **Best Poster from a Postdoctoral Scholar**, *NIF Users Group Meeting*, awarded for "X-ray imaging and electron temperature evolution in laser-driven magnetic reconnection experiments at the National Ignition Facility".
- 2024 **Best Seminar Prize**, *Princeton University Postdoctoral Council*, awarded for "Rotating Plasmas, Accretion Disks, and Jets: From the Laboratory to the Universe" as the best talk at the Postdoctoral Seminar Series.
- 2022 **Best Talk Prize**, *International Conference on High-Energy-Density Laboratory Astrophysics (HEDLA)*, awarded for "Characterization of quasi-Keplerian, differentially rotating, free-boundary laboratory plasmas".
- 2021 **Postgraduate Research Symposium Prize**, *Imperial College Department of Physics*, awarded for "Structure and dynamics of pulsed-power driven differentially rotating plasmas".

Scholarships

- 2018–2022 **Imperial College President's Ph. D. Scholarship**.
- 2016–2018 **Pontificia Universidad Catolica de Chile M. Sc. Studentship**.

Travel grants

- 2024–2025 **NNSA Facility Access Program Travel Support** for up to three co-investigators per year for project *Magnetic Ω -Effect in Differentially Rotating Plasma Flows*. (\$12k).
- 2024 **NIF Users Group Meeting**, Livermore, USA. (\$2k).
- 2023 **LaserNET US meeting**, Washington DC, USA. (\$2k)
- 2023 **OMEGA Laser Users Group (OLUG) meeting**, Laboratory for University of Rochester, USA. (\$1.5k).
- 2019 **High-Energy-Density Science Summer School**, UCSD, USA.
- 2016 **Joint IAEA-ICTP College on Plasma Physics**, ICTP, Italy.
- 2016 **Annual Chilean Physics Symposium**, Pontificia Universidad Catolica de Chile, Chile.

Selected Invited Talks

- Apr. 2026 **International Conference on High Energy Density Laboratory Astrophysics (HEDLA)**, Yokohama, Japan.
- Sept. 2025 **Plasma Science and Fusion Center Seminar**, *Massachusetts Institute of Technology*, Cambridge MA, USA.
- Apr. 2025 **Twin Talk seminar**, *Princeton Plasma Physics Laboratory*, Princeton NJ, USA.
- Oct. 2024 **APS Division of Plasma Physics Annual Meeting**, Atlanta GA, USA.
- Jul. 2024 **Space, Plasma, and Climate Research Seminar**, *Imperial College London*, London, UK.

- Jun 2024 **Plasma Astrophysics Seminar**, *Canadian Institute for Theoretical Astrophysics*, University of Toronto, Toronto, Canada.
- May 2024 **Plasma Seminars**, *University of California*, Los Angeles CA, USA.
- May 2024 **International Conference on High Energy Density Laboratory Astrophysics (HEDLA)**, Tallahassee FL, USA.
- Aug. 2023 **Optical Imaging Congress**, Optica (formerly Optical Society of America), Boston MA, USA.
- Jul. 2023 **International Conference on Dense Z-Pinches**, Ann Arbor MI, USA.
- Mar. 2023 **Astrophysics Seminar**, *Institute for Advanced Study*, Princeton NJ, USA.
- Jun. 2022 **EPS Plasma Physics Division Conference**, Virtual.
- May 2022 **International Conference on High Energy Density Laboratory Astrophysics (HEDLA)**, Lisbon, Portugal.
- Sept. 2021 **Faculty of Natural Science Research Showcase**, *Imperial College London*, London, UK.
- Nov. 2021 **Plasma Science and Fusion Center Seminar**, *Massachusetts Institute of Technology*, Cambridge MA, USA.
- Aug. 2021 **EAPPC, BEAMS & MEGAGAUSS Conference**, Biarritz, France.
- Aug. 2020 **Plasma Physics Seminar**, *University of Warwick*, UK.

Supervision

Graduate research supervision

- 2026 **Summer Intern Advisor**, Mr. Nikhil Keer, *Ph. D. student*, Plasma Science and Fusion Center, Massachusetts Institute of Technology.
- 2026–Present **Co-Advisor**, Mr. Rayner Sutardja, *Ph. D. student*, Department of Physics and Astronomy, University of California, Los Angeles, USA. *Advisor: Prof. Derek Schaeffer*.
- 2026 **Co-Advisor**, Mr. Bradyn Klein, *Postbac researcher*, Department of Physics and Astronomy, University of California, Los Angeles, USA. (Moving to U. of Maryland in Fall 2026). *Advisor: Prof. Derek Schaeffer*.

Undergraduate research supervision

- 2020 **Supervisor**. Year 1 undergraduate experimental physics, 2 projects. Department of Physics, Imperial College London, UK. (*4 students, 12 hours total*)

Academic Service

Reviewing and proposal evaluation

- 2026 **Reviewer**, ASCR Leadership Computing Challenge (ALCC), Department of Energy Advanced Scientific Computing Research Program, USA.
- 2025 **Laboratory Basic Science (LBS) Proposal Reviewer**, Laboratory for Laser Energetics, University of Rochester, USA.
- From 2023 **Peer-Reviewer**: *Physical Review Letters*, *The Astrophysical Journal*, *Review of Scientific Instruments*, *Journal of Plasma Physics*, and *High-Power Laser Science and Engineering*.

Conference and community service

- 2025 **Chair**, Session on Magnetized High-Energy Density Plasmas, APS Division of Plasma Physics Annual Meeting, Long Beach, CA, USA.
- 2024 **Program Committee**, LaserNet US Meeting, Austin, TX, USA.
- 2024 **Speaker Host**, Joint Princeton University and Institute for Advanced Study Astrophysics Colloquium Series.

Advisory and mentoring service

- 2026–Present **Ph. D. Dissertation Committee**, Mr. Lucas A. Lopez, Instituto de Fisica, Pontificia Universidad Catolica de Chile.
- 2026–Present **Ph. D. Dissertation Committee**, Ms. Consuelo B. Mimica, Instituto de Fisica, Pontificia Universidad Catolica de Chile.
- 2024–Present **Ph. D. Dissertation Committee**, Mr. Chung Hei Leung, Department of Physics and Astronomy, University of Delaware, USA.

- 2023–2024 **Mentor** to one graduate student in the Department of Astrophysical Sciences, Princeton University, USA.
- 2023 **Conference Ally Mentor**, three undergraduate students, LaserNet US Meeting, Washington, DC, USA.

Departmental service

- 2024–2025 **Climate Committee**, Department of Astrophysical Sciences, Princeton University, USA.
- 2023–2025 **Founding Member**, Astro Postdoctoral Council, Department of Astrophysical Sciences, Princeton University, USA.
- 2023–2025 **Graduate Student Liaison**, Department of Astrophysical Sciences, Princeton University, USA.
- 2013 **President**, Student Council, Department of Physics and Astronomy, Pontificia Universidad Catolica de Chile.

Teaching Experience

Classroom teaching

- Jun. 2025 **Invited Lecturer**. "Laboratory Plasma Astrophysics: Theory and Selected Topics". *Introduction to Plasmas and Fusion Course*, Princeton Plasma Physics Laboratory (1.5 hours).
- 2024–2025 **Lecturer**. *Undergraduate Summer Research Program Seminars*, Princeton University (2 hours total).
- 2017 **Teaching Assistant**. *Lagrangian and Hamiltonian Mechanics*. Pontificia Universidad Catolica de Chile, Chile. (24 hours total)
- 2016 **Teaching Assistant**. *Special and General Theory of Relativity*. Pontificia Universidad Catolica de Chile, Chile. (24 hours total)
- 2016 **Teaching Assistant**. *Modern Physics*. Pontificia Universidad Catolica de Chile, Chile. (24 hours total)
- 2014 **Teaching Assistant**. *Electromagnetism*. Pontificia Universidad Catolica de Chile, Chile. (24 hours total)
- 2014,2018 **Teaching Assistant**. *Newtonian Mechanics*. Pontificia Universidad Catolica de Chile, Chile. (48 hours total)
- 2014–2017 **Teaching Assistant**. *Physics for Biological and Chemical Sciences*. Pontificia Universidad Catolica de Chile, Chile. (72 hours total)
- 2013,2017 **Teaching Assistant**. *Physics for Education Students*. Pontificia Universidad Catolica de Chile, Chile. (48 hours total)

Laboratory teaching

- 2018–2021 **Demonstrator**. Year 2 Physics *Waves and Their Propagation* undergraduate lab. Including remote demonstrator with **Lab-In-A-Box** in 2020. Department of Physics, Imperial College London, UK. (180 hours total)

Selected Conference Contributions

- Nov. 2025 APS Division of Plasma Physics Annual Meeting, Long Beach CA, USA. (Talk)
- Mar. 2025 International Workshop of Supernova Remnant Physics in the Lab, Lorentz Center, Leiden, Holland. (Talk)
- Jan. 2025 Latin American Workshop on Plasma Physics, Santiago, Chile (Talk)
- Oct. 2024 APS Division of Plasma Physics Annual Meeting, Atlanta GA, USA. (Poster and Invited Talk)
- Aug. 2024 International Workshop on the Interrelationship between Plasma Experiments in the Laboratory and in Space (IPELS-16), Garching, Germany. (Poster)
- Jul. 2024 EPS Plasma Physics Division Conference, Salamanca, Spain. (Talk)
- May 2024 International Conference on High Energy Density Laboratory Astrophysics (HEDLA), Tallahassee FL, USA. (Poster and Invited Talk)
- Feb. 2024 National Ignition Facility Users Group Meeting, Livermore CA, USA. (Poster)
- Nov. 2023 APS Division of Plasma Physics Annual Meeting, Denver CO, USA. (Talk)
- Nov. 2022 APS Division of Plasma Physics Annual Meeting, Spokane WA, USA. (Talk)
- Nov. 2021 APS Division of Plasma Physics Annual Meeting, Pittsburgh PA, USA. (Talk)

- Apr. 2021 Plasma Physics Conference, Institute of Physics, Virtual. (Talk)
 Nov. 2020 APS Division of Plasma Physics Annual Meeting, Virtual. (Talk)
 Nov. 2019 APS Division of Plasma Physics Annual Meeting, Fort Lauderdale FL, USA. (Talk)
 Jul. 2019 HEDS Summer School, San Diego CA, USA. (Poster)
 Nov. 2018 APS Division of Plasma Physics Annual Meeting, Portland OR, USA. (Poster)
 Nov. 2016 Joint IAEA-ICTP College on Plasma Physics, Trieste, Italy. (Poster)
 Nov. 2016 Chilean Physics Annual Symposium, Santiago, Chile. (Poster)
 Oct. 2014 National Physics and Astronomy Student Symposium, Valparaiso, Chile. (Talk)

Publications

Twenty peer-reviewed articles, five as first or corresponding author.

20. 2026 K. Marrow, J. Strucka, S. Merlini, L. G. Suttle, T. Mundy, A. Uras, S. N. Bland, **V. Valenzuela-Villaseca**, S. V. Lebedev, “Measuring dynamics of differentially rotating, unmagnetized, free-boundary plasma produced by soft X-ray irradiation”, *Plasma Physics and Controlled Fusion*, 68, 025027. <https://iopscience.iop.org/article/10.1088/1361-6587/ae44cc>
19. 2026 L. Izquierdo, F. Veloso, M. Escalona, **V. Valenzuela-Villaseca**, G. Avaria, J. C. Valenzuela, “Influence of conical wire array geometry on plasma flow and temperature profiles of radiatively-cooled jets”, *Physical Review E* 113, 025203. <https://doi.org/10.1103/pvcm-kgvw>
18. 2025 B. P. McCluskey, J. Griff-McMahon, D. Haberberger, **V. Valenzuela-Villaseca**, H. Landsberger, W. Fox, “Imaging of plasma density using angular filter refractometry and a fast marching Eikonal solver”, *Review of Scientific Instruments*, 96, 123502. <https://doi.org/10.1063/5.0295850>
17. 2025 L. Orusa and **V. Valenzuela-Villaseca** (*corresponding author*), “Criteria for ion acceleration in laboratory perpendicular magnetized collisionless shocks: when are 2D simulations enough?”, *Physics of Plasmas*, 32, 052901. <https://doi.org/10.1063/5.0269035>
16. 2024 J. Griff-McMahon, **V. Valenzuela-Villaseca**, S. Malko, G. Fiksel, M. J. Rosenberg, D. B. Schaeffer, W. Fox, “Proton radiography inversions with source extraction and comparison to mesh methods”, *Physical Review E*, 110, 055202. <https://doi.org/10.1103/PhysRevE.110.055202>
15. 2024 **V. Valenzuela-Villaseca**, L. G. Suttle, F. Suzuki-Vidal, J. W. D. Halliday, S. Merlini, D. R. Russell, E. R. Tubman, J. D. Hare, J. P. Chittenden, M. E. Koepke, S. V. Lebedev, “Structure and dynamics of magneto-inertial, differentially rotating laboratory plasmas”. *Journal of Plasma Physics*, 90, 915900401. <https://doi.org/10.1017/S0022377824000710>
14. 2024 **V. Valenzuela-Villaseca**, J. M. Molina, D. B. Schaeffer, S. Malko, J. Griff-McMahon, K. Lezhnin, M. J. Rosenberg, S. X. Hu, D. Kalantar, C. Trosseille, H.-S. Park, B. A. Remington, G. Fiksel, D. Uzdensky, W. Fox, “X-ray imaging and electron temperature evolution in laser-driven magnetic reconnection experiments at the National Ignition Facility”, *Physics of Plasmas*, 31, 082106. <https://doi.org/10.1063/5.0213598>
13. 2024 J. Griff-McMahon, S. Malko, **V. Valenzuela-Villaseca**, C. A. Walsh, G. Fiksel, M. J. Rosenberg, D. B. Schaeffer, W. Fox, “Measurements of extended magnetic fields in laser-solid interactions”, *Physical Review Research*, 6, 033312. <https://doi.org/10.1103/PhysRevResearch.6.033312>
12. 2024 S. Malko, D. B. Schaeffer, W. Yao, **V. Valenzuela-Villaseca**, C. Johnson, G. Fiksel, A. Ciardi, W. Fox, “Observation of a magneto-Rayleigh-Taylor instability in magnetically collimated plasma jets”, *Physical Review Research*, 6, 023330. <https://doi.org/10.1103/PhysRevResearch.6.023330>
11. 2024 **V. Valenzuela-Villaseca**, L. G. Suttle, F. Suzuki-Vidal, J. W. D. Halliday, S. Merlini, D. R. Russell, E. R. Tubman, J. D. Hare, J. P. Chittenden, M. E. Koepke, S. V. Lebedev, “On the structure of plasma jets in the Rotating Plasma Experiment”, *IEEE Transactions on Plasma Science*, <https://doi.org/10.1109/TPS.2024.3387304>
10. 2023 **V. Valenzuela-Villaseca**, L. G. Suttle, F. Suzuki-Vidal, J. W. D. Halliday, S. Merlini, D. R. Russell, E. R. Tubman, J. D. Hare, J. P. Chittenden, M. E. Koepke, S. V. Lebedev, “Characterization of quasi-Keplerian, differentially rotating, free-boundary laboratory plasmas”, *Physical Review Letters*, 130, 195101.
Editor’s Suggestion. <https://doi.org/10.1103/PhysRevLett.130.195101>.
Featured in Physics Magazine: <https://physics.aps.org/articles/v16/81>.

9. 2023 D. R. Russell, G. C. Burdiak, J. J. Carroll-Nellenback, J. W. D. Halliday, J. D. Hare, S. Merlini, L. G. Suttle, **V. Valenzuela-Villaseca**, S. J. Eardley, J. A. Fullalove, G. C. Rowland, R. A. Smith, A. Frank, P. Hartigan, A. L. Velikovich, S. V. Lebedev, “Observation of subcritical shocks in a collisional laboratory plasma: scale dependence near the resistive length”, *Journal of Plasma Physics* 84, 915890401. <https://doi.org/10.1017/S0022377823000740>
8. 2023 S. Merlini, J. D. Hare, G. C. Burdiak, J. W. D. Halliday, A. Ciardi, J. P. Chittenden, T. Clayson, A. J. Crilly, S. J. Eardley, K. Marrow, D. R. Russell, R. A. Smith, L. G. Suttle, E. R. Tubman, **V. Valenzuela-Villaseca**, T. W. O. Varnish, S. V. Lebedev, “Radiative cooling effects on reverse shocks formed by magnetised supersonic plasma flows”, *Physics of Plasmas* 30, 092102. <https://doi.org/10.1063/5.0160809>
7. 2022 D. R. Russell, G. C. Burdiak, J. J. Carroll-Nellenback, J. W. D. Halliday, J. D. Hare, S. Merlini, L. G. Suttle, **V. Valenzuela-Villaseca**, S. J. Eardley, J. A. Fullalove, G. C. Rowland, R. A. Smith, A. Frank, P. Hartigan, A. L. Velikovich, S. V. Lebedev, “Perpendicular Subcritical Shock Structure in a Collisional Plasma Experiment”, *Physical Review Letters* 129, 225001 <https://doi.org/10.1103/PhysRevLett.129.225001>
6. 2022 J. W. D. Halliday, A. Crilly, J. P. Chittenden, R. Mancini, S. Merlini, S. Rose, D. R. Russell, L. G. Suttle, **V. Valenzuela-Villaseca**, S. N. Bland, S. V. Lebedev, “Investigating Radiatively Driven, Magnetised Plasmas with a University Scale Pulsed-Power Generator”, *Physics of Plasmas*, 29, 042107. <https://doi.org/10.1063/5.0084550>
5. 2021 L. G. Suttle, J. D. Hare, J. W. D. Halliday, S. Merlini, D. R. Russell, E. R. Tubman, **V. Valenzuela-Villaseca**, W. Rozmus, C. Bruulsema, S. V. Lebedev, “Collective optical Thomson scattering in pulsed-power driven high energy density physics experiments”, *Review of Scientific Instruments*, 92, 033542. <https://doi.org/10.1063/5.0041118>
4. 2020 F. Veloso, G. Muñoz-Cordovez, D. Diaz-Droguett, **V. Valenzuela-Villaseca**, M. Vescovi, H. Bhuyan, M. Favre, “Axial outflows from conical wire array z-pinches as a tool for surface modifications”, *Results in Physics*, 19, 103528. <https://doi.org/10.1016/j.rinp.2020.103528>
3. 2018 G. Muñoz-Cordovez, F. Veloso, **V. Valenzuela-Villaseca**, M. Vescovi, W. Useche, E. Wyndham, M. Favre, “Emission of fast ions from conical wire array Z-pinches studied at different background pressures”, *Physics of Plasmas*, 25, 102101. <https://doi.org/10.1063/1.5045215>
2. 2016 M. P. Valdivia, D. Stutman, C. Stoeckl, C. Mileham, I. Begishev, W. Theobald, J. Bromage, S. P. Regan, S. R. Klein, G. Muñoz-Cordovez, M. Vescovi, **V. Valenzuela-Villaseca**, F. Veloso, “Talbot-Lau x-ray deflectometer electron density diagnostic for laser and pulsed power high energy density plasma experiments”, *Review of Scientific Instruments*, 87, 11D501. <https://doi.org/10.1063/1.4959158>
1. 2015 F. Veloso, G. Muñoz-Cordovez, L. Donoso-Tapia, **V. Valenzuela-Villaseca**, F. Suzuki-Vidal, G. F. Swadling, J. P. Chittenden, M. Favre, E. Wyndham, “Ablation dynamics in wire array Z-pinches under modifications on global magnetic field topology”, *Physics of Plasmas*, 22, 072509. <https://doi.org/10.1063/1.4926581>

Preprints in Review

Six articles in review, including one first-author manuscript.

- Submitted **V. Valenzuela-Villaseca**, S. Totorica, J. Griff-McMahon, L.-J. Chen, S. Malko, P. V. Heuer, P. Pongkitiwanichakul, W. Fox, D. B. Schaeffer, “Laboratory measurements of energy partitioning and anomalous electron heating in magnetized, perpendicular collisionless shocks”. In review at *The Astrophysical Journal*. Preprint: <https://arxiv.org/abs/2509.12164v2>
- Submitted J. Griff-McMahon, C. A. Walsh, **V. Valenzuela-Villaseca**, S. Malko, B. P. McCluskey, K. Lezhnin, H. Landsberger, G. Fiksel, M. J. Rosenberg, D. B. Schaeffer, W. Fox, “Structure of self-generated magnetic fields in laser-solid interaction”. In review at *Physical Review Letters*. Preprint: <https://arxiv.org/abs/2510.14076>
- Submitted J. Griff-McMahon, **V. Valenzuela-Villaseca**, C. A. Walsh, S. Malko, B. P. McCluskey, K. Lezhnin, H. Landsberger, G. Fiksel, M. J. Rosenberg, D. B. Schaeffer, W. Fox, “Evolution of laser-driven magnetic fields from proton tomography”. In review at *Physics of Plasmas*. Preprint: <https://arxiv.org/pdf/2603.18931>
- Submitted G. D. Sutcliffe, N. Vanderloo, C. Bruulsema, **V. Valenzuela-Villaseca**, G. F. Swadling, M. Zhou, B. B. Pollock, A. Bret, J. S. Ross, J. Moody, C. K. Li, “Sustained interpenetrating plasma flows for the investigation of late time kinetic instability evolution”. In review at *Physics of Plasmas*. Preprint: <https://arxiv.org/abs/2605.10773>

- Submitted S. Merlini, J. R. Beattie, **V. Valenzuela-Villasaca**, “Numerical simulations of shock-driven, supersonic turbulence in colliding three-temperature laboratory plasmas”. In review at *The Astrophysical Journal*. Preprint: <https://arxiv.org/abs/2605.21672>
- Submitted S. Malko, K. Bhutwala, X. Vaisseau, V. Ospina-Bohorquez, R. Nedbailo, J. Griff-McMahon, W. Cayzac, J. I. Apiñaniz, M. Ehret, A. Huerta, D. De Luis Blanco, M. Bailly-Grandvaux, J. Vargas, C. McGuffey, **V. Valenzuela-Villasaca**, C. Johnson, A. Essa, R. Hollinger, S. Wang, G. Zeraouli, M. Huault, P. Neumayer, J. A. Perez Hernandez, K. Lezhnin, F. Kraus, W. Fox, A. Tauschwitz, J. J. Santos, F. N. Beg, J. Rocca, S. Atzeni, R. Fedosejevs, L. Volpe, “Platform for high repetition rate proton stopping power measurements in Warm Dense Matter”. In review at *Plasma Physics and Controlled Fusion*.

In the Press & Media

I have appeared in some news outlets, both in English and Spanish. Selected examples are listed below.

- Oct. 2024 **Laboratory for Laser Energetics**. Quick Shot article *OMEGA 60-Beam Laser System Helps Shed Light onto Black Hole Accretion Disks*.
- Oct. 2024 **American Physical Society, Division of Plasma Physics**. Press release *Closer to the Horizon: Laser and Pulsed-Power Experiments Emulate Black Hole Accretion Disks*.
- Sep. 2023 **Scientific American**. Comment on article *Supermassive Black Hole Feeding Frenzies May Explain Blinking Quasars* by Stephanie Pappas.
- Oct. 2023 **The Times**. Related article *Blast off simulator in London recreating the birth of stars* by Niall Jeffrey.
- May 2023 **Imperial College London**. Press Release *Shining ring around black holes recreated in the lab*, by Hayley Dunning.
- May 2023 **Tendencias 21**. Featured article *Se puede observar lo que pasa en el universo sin salir de casa* by Eduardo Martinez.

Collaborators

The breadth of my research interests has supported a number of fruitful collaborations, with my ongoing partnerships listed below. Graduate students are highlighted with an asterisk (*) and postdocs with a double asterisk (**) below.

- **Chris Walsh** (LLNL), **George Swadling** (LLNL), **Eric Blackman** (Rochester), and **Nikhil Keer*** (MIT): Magnetorotational instability and magnetized rotating winds.
- **Katherine Marrow*** (Imperial): X-ray driven rotating plasmas.
- **Nick Schwartz**** and **Johan Frenje** (MIT): Next generation pulsed-power.
- **Stefano Merlini**** (Imperial) and **James Beattie**** (CITA/Princeton): Plasma turbulence.
- **Derek Schaeffer** (UCLA), **Luca Orusa**** (Columbia/Princeton), and **Julien Fuchs** (CNRS): Collisionless shocks.
- **Will Fox** (Maryland), **Jesse Griff-McMahon*** (Princeton), and **Huws Landsberger*** (Princeton): Self-generation of magnetic fields, reconnection, and transport in high-beta laser-plasmas.
- **Graeme Sutcliffe**** (LLNL) and **Niels Vanderloo*** (MIT): Nonlinear Weibel instability and kinetic turbulence.
- **Sophia Malko** (PPPL): Nernst transport in magneto-inertial fusion plasmas.
- **Lee Suttle** (Imperial), **Sergey Lebedev** (Imperial), **Sam Totorica** (Maryland), **Frederico Fiuza** (IST), and **Jack Halliday** (RAL): Particle acceleration in pulsed-power magnetic reconnection experiments.
- **Skylar Dannhoff*** and **Chikang Li** (MIT): Hohlraum plasma physics experiments.
- **Luisa Izquierdo**** (Cornell): Plasma jets and shocks.
- **Patrick Knapp** (Pacific Fusion), **Brian O'Shea** (Michigan State), and **Hannah Hasson**** (SNL): MHD plasma jets on Z.